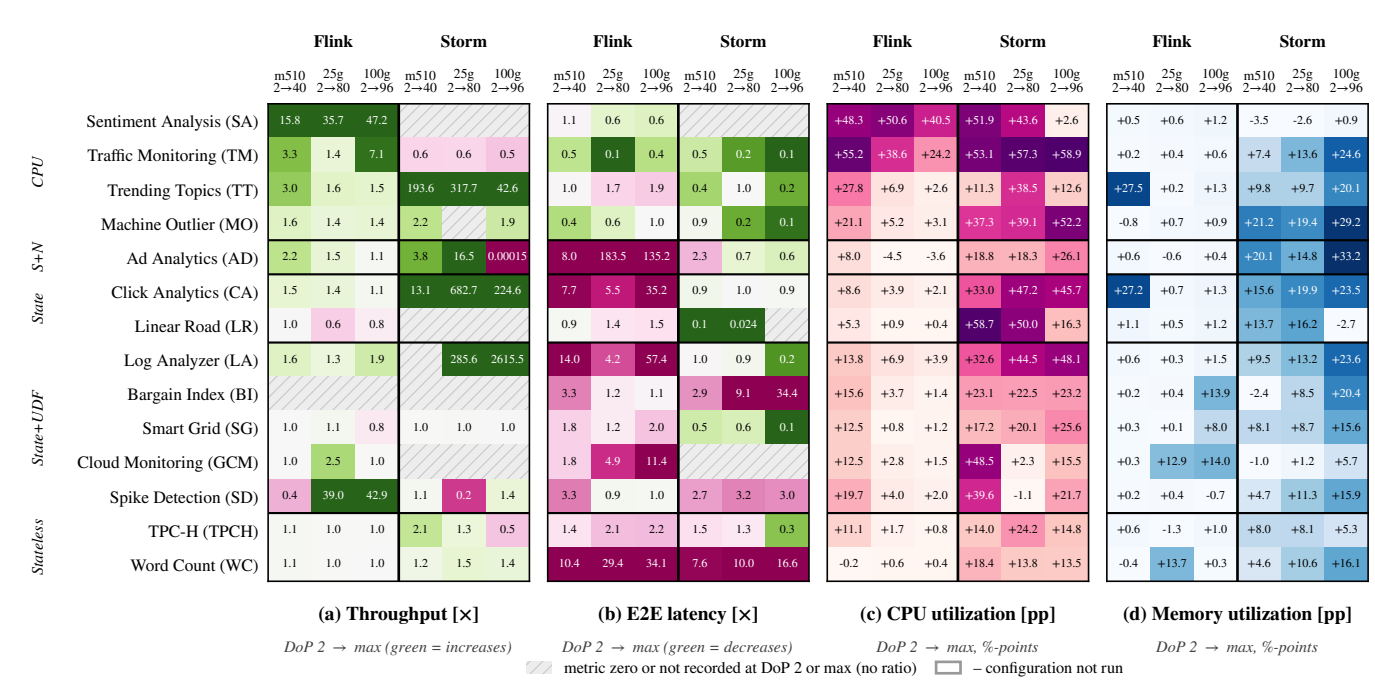
Flink
Storm
m510 2→40
25g 2→80
100g 2→96
CPU
S+N
State
State+UDF
Stateless
Sentiment Analysis (SA) 15.8 35.7 47.2 | 1.1 0.6 0.6 | +48.3 +50.6 +40.5 | +51.9 +43.6 +2.6 | +0.5 +0.6 +1.2 | -3.5 -2.6 +0.9
Traffic Monitoring (TM) 3.3 1.4 7.1 | 0.6 0.6 0.5 | 0.5 0.1 0.4 | 0.5 0.2 0.1 | +55.2 +38.6 +24.2 | +53.1 +57.3 +58.9 | +0.2 +0.4 +0.6 | +7.4 +13.6 +24.6
Trending Topics (TT) 3.0 1.6 1.5 | 193.6 317.7 42.6 | 1.0 1.7 1.9 | 0.4 1.0 0.2 | +27.8 +6.9 +2.6 | +11.3 +38.5 +12.6 | +27.5 +0.2 +1.3 | +9.8 +9.7 +20.1
Machine Outlier (MO) 1.6 1.4 1.4 | 2.2 1.9 | 0.4 0.6 1.0 | 0.9 0.2 0.1 | +21.1 +5.2 +3.1 | +37.3 +39.1 +52.2 | -0.8 +0.7 +0.9 | +21.2 +19.4 +29.2
Ad Analytics (AD) 2.2 1.5 1.1 | 3.8 16.5 0.00015 | 8.0 183.5 135.2 | 2.3 0.7 0.6 | +8.0 -4.5 -3.6 | +18.8 +18.3 +26.1 | +0.6 -0.6 +0.4 | +20.1 +14.8 +33.2
Click Analytics (CA) 1.5 1.4 1.1 | 13.1 682.7 224.6 | 7.7 5.5 35.2 | 0.9 1.0 0.9 | +8.6 +3.9 +2.1 | +33.0 +47.2 +45.7 | +27.2 +0.7 +1.3 | +15.6 +19.9 +23.5
Linear Road (LR) 1.0 0.6 0.8 | 0.9 1.4 1.5 | 0.1 0.024 | +5.3 +0.9 +0.4 | +58.7 +50.0 +16.3 | +1.1 +0.5 +1.2 | +13.7 +16.2 -2.7
Log Analyzer (LA) 1.6 1.3 1.9 | 285.6 2615.5 | 14.0 4.2 57.4 | 1.0 0.9 0.2 | +13.8 +6.9 +3.9 | +32.6 +44.5 +48.1 | +0.6 +0.3 +1.5 | +9.5 +13.2 +23.6
Bargain Index (BI) | 3.3 1.2 1.1 | 2.9 9.1 34.4 | +15.6 +3.7 +1.4 | +23.1 +22.5 +23.2 | +0.2 +0.4 +13.9 | -2.4 +8.5 +20.4
Smart Grid (SG) 1.0 1.1 0.8 | 1.0 1.0 1.0 | 1.8 1.2 2.0 | 0.5 0.6 0.1 | +12.5 +0.8 +1.2 | +17.2 +20.1 +25.6 | +0.3 +0.1 +8.0 | +8.1 +8.7 +15.6
Cloud Monitoring (GCM) 1.0 2.5 1.0 | 1.8 4.9 11.4 | +12.5 +2.8 +1.5 | +48.5 +2.3 +15.5 | +0.3 +12.9 +14.0 | -1.0 +1.2 +5.7
Spike Detection (SD) 0.4 39.0 42.9 | 1.1 0.2 1.4 | 3.3 0.9 1.0 | 2.7 3.2 3.0 | +19.7 +4.0 +2.0 | +39.6 -1.1 +21.7 | +0.2 +0.4 -0.7 | +4.7 +11.3 +15.9
TPC-H (TPCH) 1.1 1.0 1.0 | 2.1 1.3 0.5 | 1.4 2.1 2.2 | 1.5 1.3 0.3 | +11.1 +1.7 +0.8 | +14.0 +24.2 +14.8 | +0.6 -1.3 +1.0 | +8.0 +8.1 +5.3
Word Count (WC) 1.1 1.0 1.0 | 1.2 1.5 1.4 | 10.4 29.4 34.1 | 7.6 10.0 16.6 | -0.2 +0.6 +0.4 | +18.4 +13.8 +13.5 | -0.4 +13.7 +0.3 | +4.6 +10.6 +16.1
(a) Throughput [×]
(b) E2E latency [×]
(c) CPU utilization [pp]
(d) Memory utilization [pp]
DoP 2 → max (green = increases)
DoP 2 → max (green = decreases)
DoP 2 → max, %-points
DoP 2 → max, %-points
metric zero or not recorded at DoP 2 or max (no ratio)
– configuration not run